

HAREENA CHOWDARY POLAVARAM

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SKILLS

Cloud Platforms: AWS (S3, EC2, Lambda, Redshift, Glue), Azure (Azure Data Factory, Databricks, Data Lake, Synapse), GCP (BigQuery)

Big Data & ETL Tools: PySpark, SparkSQL, Delta Lake, Airflow, Kafka, Azure Data Factory, dbt, Informatica (PowerCenter, BDM)

DevOps & CI/CD: Azure DevOps, Jenkins, Docker, Kubernetes, Terraform

Languages & Scripting: Python, SQL, T-SQL, HTML, CSS, JavaScript

Databases: MySQL, SQL Server, PostgreSQL, Oracle, DynamoDB

Visualization & Tools: Power BI, Excel VBA, Git, Jupyter, VS Code, JIRA, Grafana, Splunk

EXPERIENCE

Data Operations Assistant | Western Michigan University, USA

Sep 2023 – Apr 2025

- Analyzed institutional records in PeopleSoft using SQL queries to identify trends that guided academic resource allocation and departmental decision-making.
- Increased request fulfillment efficiency by 30% by streamlining workflows with cross-functional teams and external stakeholders.
- Maintained 100% FERPA compliance by validating schemas and enforcing data governance across student record systems.
- Engineered weekly reporting workflows using Python and Excel, saving approximately 12 hours per month in manual effort.
- Investigated transformation errors in student data flows, improving report accuracy by 20% and enhancing operational reliability.

Data Engineer | Tata Consultancy Services, India

Aug 2021 – Jul 2023

- Built and maintained scalable ETL pipelines using Informatica PowerCenter, Informatica BDM, Azure Data Factory, and Databricks to ingest, transform, and load multi-source data for enterprise reporting.
- Optimized SQL queries and transformations in Oracle and Databricks, reducing execution time by ~25% and improving pipeline efficiency.
- Developed complex mappings and datasets using transformations such as Lookup, Joiner, Aggregator, Router, Rank, and Source Qualifier to meet analytical requirements.
- Validated and cleansed data through automated checks in ADF and SQL, achieving 99% data accuracy across environments.
- Migrated ETL mappings, datasets, and workflows across development, QA, and production environments, ensuring smooth deployments with minimal downtime.
- Translated business requirements into technical STTM documents and mapping logic, enhancing traceability and reducing rework by 15%.
- Resolved production defects and data issues reported by clients, maintaining 100% SLA compliance and supporting data quality initiatives.
- Tracked project tasks, progress, and issue resolution using JIRA and Azure DevOps while collaborating with cross-functional teams on cloud migration and data transformation projects.

AI Intern | National Instruments (Cognibot), Remote

May 2020 – Jun 2020

- Constructed and deployed ML models for predictive maintenance using 100K+ IoT industrial telemetry points from sensors and devices; achieved >85% anomaly detection accuracy across test environments.
- Integrated real-time edge analytics to trigger alerts with sub-2s latency, improving responsiveness to mechanical faults.
- Enhanced model pipelines through hyperparameter tuning and feature engineering, increasing overall F1 score by 18%.

EDUCATION

Western Michigan University

Master of Science in Computer Science

Kalamazoo, MI | Aug 2023 – Apr 2025

GPA: 3.7/4

Sri Venkateswara University

Bachelor of Technology in Computer Science

Tirupati, India | Jul 2017 – Jul 2021

GPA: 3.6/4

Coursework: Machine Learning, AI, Big Data Analysis using Python, Algorithms, Advanced Databases Concepts, Statistics

PROJECTS

Azure Databricks End-to-End Data Engineering Pipeline

- Engineered an end-to-end data pipeline on Azure Databricks, leveraging PySpark and Auto Loader for the incremental ingestion of raw data from Azure Data Lake Storage (ADLS) Gen2 into a multi-layered Medallion architecture (Bronze, Silver, Gold).
- Developed a dimensional data warehouse by implementing a Star Schema with Slowly Changing Dimensions (SCD Type 1 & 2) using Delta Live Tables (DLT), and managed data governance, access control, and lineage through Unity Catalog.

Azure End-to-End Data Engineering Project

- Engineered a production-ready data pipeline using Azure Data Factory for incremental ingestion from Azure SQL and utilized Azure Databricks to transform and model the data into a star schema within a Delta Lakehouse.
- Implemented a robust data governance framework using Unity Catalog for fine-grained access control and automated the entire multi-layered workflow with Databricks Workflows, optimizing job execution by running tasks in parallel.

AdventureWorks Data Engineering Project using Azure Stack

- Designed and deployed a data engineering pipeline on Azure (Data Lake, Data Factory, Databricks, Synapse, Power BI) to ingest, transform, and model AdventureWorks sales and customer data for advanced analytics.
- Implemented ETL transformations with PySpark to build curated data layers in Parquet, applied data quality checks, and delivered optimized, BI-ready datasets for interactive reporting in Power BI.

F1 Race Data Engineering Project

- Created a scalable F1 race analytics lakehouse by ingesting Ergast API data into Azure Data Lake, transforming it with PySpark in Databricks, and storing with ACID compliance in Delta Lake.
- Automated 20+ ETL workflows via Azure Data Factory and delivered Power BI dashboards covering 1,000+ races and 70+ circuits for race insights and historical comparisons.